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Whatever it takes to understand a central banker — Embedding their w using neural networks☆

Martin Baumgärtner [a,b], Johannes Zahner [c,d,*]

[a] THM Business School, Germany
[b] Justus-Liebig-University Gießen, Germany
[c] Goethe University Frankfurt, Germany
[d] Marburg Centre for Institutional Economics, Germany

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ABSTRACT

Dictionary-based methods represent the most commonly used approach for quantifying the qualitative information from (central bank) communication. In this paper, we propose machine learning models that generates embeddings from words and documents. Embeddings are multi-dimensional numerical text representations that capture the underlying semantic relationships within text. Using a novel corpus of 22,000 documents from 128 central banks, we generate the first domain-specific embeddings for central bank communication that outperform dictionaries and existing embeddings on tasks such as predicting monetary policy shocks. We further demonstrate the efficacy of our embeddings by constructing an index that tracks the extent to which Federal Reserve communications align with an inflation-targeting stance. Our empirical results indicate that deviations from inflation-targeting language substantially affect market-based expectations and influence monetary policy decisions, significantly reducing the inflation response parameter in an estimated Taylor rule.

1. Introduction

What did ECB president Mario Draghi mean on July 26, 2012, when he stated that *"within [its] mandate, the ECB is ready to do whatever it takes to preserve the euro"*? According to most commonly used dictionary-based methods this as a neutral sentence, due to the lack of positive or negative terms. However, the message contained was nothing short of extraordinary for financial market participants and monetary policy experts; in fact, it marked a turning point in the ongoing euro area crisis. In this paper, we propose a novel method capable of capturing such subtleties.

Over the past two decades, there has been an increase in the use of unstructured datasets in monetary policy, in particular concerning the analysis and interpretation of central bank communication (Blinder et al., 2008). The development was accelerated by

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* Corresponding author at: Goethe University Frankfurt, Germany.
E-mail address: johannes.zahner@gmail.com (J. Zahner).

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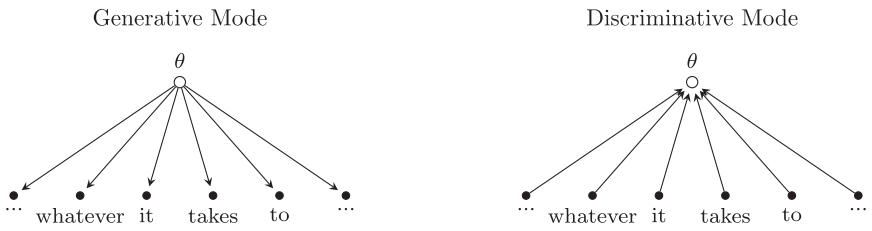


Fig. 1. Communication model.
Note: The illustration is adapted from Lowe (2021, p. 10).

the zero lower bound and the emergence of forward guidance, wherein central bankers recognized the possibility of complementing actions with well-placed language to steer market participants towards the desired equilibrium path. As a result, central banks have substantially expanded their communication channels. In 2011, for instance, the Fed began holding regularly scheduled press conferences to explain its monetary policy decisions.

Research on such central bank communication is typically based on the presumption that it contains latent messages (θ) by the policymakers, which are worth extracting. While not observable directly, these latent messages generate variations in the choice of words (W)–a process depicted on the left-hand side of Fig. 1. Since only the outcome of this sampling process can be directly observed, it is the researchers' job to infer the underlying message from the variation in W, as illustrated by the right-hand side. This paper aims to provide an effective representation for the words that allows one to effectively retrieve the underlying messages from the observed variation in communication ($W \rightarrow \theta$).

The current literature predominantly relies on dictionary-based term counts to extract a single dimension (for instance, sentiment) from a document. While dictionaries offer a straightforward and transparent assessment, they frequently miss relevant terms, rely on subjective construction, and typically yield low-dimensional representations. Although they may suffice for simple messages, dictionaries do not capture the full richness of central bank communications and have thus been found to be dominated by noise rather than signal (Hayo and Zahner, 2023). To address these shortcomings, computational linguistics has turned to machine learning to develop novel language models that generate meaningful, multidimensional text representations–so-called *embeddings*.

In this paper, we develop a language model that is trained explicitly for central bank communication. Our focus is twofold. On the one hand, we aim to sharpen the previously broad focus of embeddings, while, on the other hand, enhance content extraction compared to dictionary approaches. As such, this paper contributes to the current literature on four fronts. First, we collect a novel text corpus of central bank communication unparalleled in size and diversity, with around 22,000 speeches by 128 central banks. Second, we introduce and evaluate several novel machine learning algorithms for text quantification. Third, we identify the best-performing algorithm and train it on our specialized corpus to produce the first domain-specific embeddings for central bank communication–outperforming dictionaries and existing embeddings on tasks such as predicting monetary policy shocks. We provide suggestive evidence that this superior performance stems from the training on a domain-specific corpus. Fourth, using our best-performing embeddings, we construct an index derived from the Fed's inter-meeting communications, which tracks the strength of its inflation-targeting regime. In two empirical exercises, we demonstrate that this index influences both inflation expectations and moderates the Fed's inflation response.

The remainder of this paper is structured as follows. Following a literature review in Section 2, we introduce both the text corpus and the machine learning algorithms in Section 3. Section 4 evaluates the quality of the resulting embeddings. The best-performing embeddings are then applied in Section 5 to develop the Fed's inflation-targeting regime index. The final section concludes the paper.

2. Related literature

> *"Approaches [...] which use embeddings as the basis for mathematical analyses of text, can play a role in the next generation of text-as-data applications in social science"*
>
> — Gentzkow et al. (2019, p. 553)

There are several methods available to researchers for quantifying qualitative information for econometric models. One (labor-intensive) approach is manual classification, whereby sentences, paragraphs, or sections are categorized by hand–often referred to as the "narrative" approach. Manual classification maximizes recall, that is, the avoidance of false negatives, by being able to capture highly specific patterns. For instance, Friedman and Schwartz (1963), Romer and Romer (2004) analyze internal Fed deliberations and debates to identify true monetary policy shocks.

In the past decade, computational methods have become prominent in economics. Gentzkow et al. (2019), Chakraborty and Joseph (2017), Ash and Hansen (2023) provide excellent surveys on the use of text data, with a focus on economics, monetary policy, and machine learning respectively. To date, most applications rely on so-called dictionaries that assign a categorical classification

to specific terms, thereby quantifying the qualitative information into few dimensions. These dictionary-based methods inherently assume that the text corpus provides limited information about the concept of interest and therefore apply strong priors regarding the specific language used to describe that concept. In many cases, these assumptions hold, making dictionary-based approaches the most appropriate tool for the analysis. Notable examples include the construction of an economic uncertainty index through term frequency counts in news articles (Baker et al., 2016), or stock market predictions using a psychosocial dictionary on a Wall Street Journal column (Tetlock, 2007).

Dictionaries have also been widely used in the context of central bank communication, some being explicitly designed for financial and monetary policy language (*e.g.*, Loughran and McDonald, 2011; Apel and Grimaldi, 2014; Bennani and Neuenkirch, 2017; Correa et al., 2021). The necessity of central bank-specific dictionaries arises from the language employed in this field. Take, for instance, the term LIABILITY, which carries a negative connotation in common parlance while it is a purely technical term within the realms of finance and monetary policy (Loughran and McDonald, 2011). Thus, the advantage of dictionary-based methods is that they are easy to implement, understand, and evaluate through their straightforward and transparent quantification of an underlying corpus. However, dictionaries also present significant limitations, including oversimplification of language, omission of potentially relevant information, and a lack of objectivity.

First, in dictionaries the relationship between concepts and words is characterized by a prior assumption that most words are irrelevant, resulting in the exclusion of a substantial portion of text. Moreover, dictionary construction is inherently subjective. Researchers curate a subset of a language's vocabulary based on their interpretation of each term's meaning, potentially introducing biases that can distort the results. Finally, terms defined in dictionaries are typically classified in binary terms (such as positive and negative or hawkish and dovish), which limits their ability to capture nuances. As a result, dictionary approaches quantify complex concepts in low-dimensional representations, prone to measurement error. For instance, the phrase GREAT RECESSION would be classified as neutral using the sentiment dictionary of Loughran and McDonald's (2011), since GREAT is labeled positive and RECESSION negative, even though GREAT in this context does not carry a positive connotation. While expanding dictionaries to include more sophisticated terms or bi-grams (*e.g.*, GREAT_RECESSION) may improve specificity, it comes at the cost of reduced sensitivity. Consequently, dictionary-based methods have been found to be extremely noisy. For instance, Hayo and Zahner (2023) examined how much variation in sentiment-based indicators of central bank communication could be attributed to changes in macroeconomic, financial, and monetary variables. The authors found that these factors explained only a small fraction of the underlying variation–typically less than 5% and conclude that their *"findings cast some doubt on the reliability of conclusions [...] that are based on variations in dictionary-based sentiment indicators"* (Hayo and Zahner, 2023, p. 5).

Recent work by Kalamara et al. (2022), Shapiro et al. (2020), Kanelis and Siklos (2022) acknowledge these limitations, suggesting augmenting indices or combining multiple dictionaries. Their findings indicate that dictionaries complement each other, highlighting different distinct dimensions of an underlying corpus. In Section 4, we replicate this approach and observe moderate improvements in predictive power when combining up to four dictionaries.

In addition to dictionary-based augmentations, alternative approaches are becoming more popular, with two prominent examples being the concepts of *similarity* and *complexity*. Similarity measures the distance between two documents' vocabulary (explored further in Section 4), whereas complexity measures the readability of a document. Examples of the former are Acosta and Meade (2015), Ehrmann and Talmi (2020), who find that introductory statements of major central banks became more similar over time. Meanwhile, Smales and Apergis (2017), Hayo et al. (2022) illustrate that markets react strongly to the complexity of the information conveyed in press statements. As helpful as these new approaches are, some of the corpus' relevant information remains neglected. For example, replacing the term INFLATION with DEFLATION in a document does not affect its measured complexity, but substantially affects the intended meaning.

In recent years, word embeddings have emerged as a powerful tool in the analysis of central bank communication, a development predicted by Gentzkow et al. (2019), as highlighted by their quote at the beginning of this section. Word embeddings are multidimensional word representations that have been applied across various domains. For instance, Apel et al. (2019) analyze central banker disagreement, Handlan (2020), Hansen and Kazinnik (2023) generate monetary policy shocks from Fed communication, Cieslak et al. (2021) improve the aforementioned uncertainty indices, Bertsch et al. (2022) measures the Fed's interpretation of its financial stability mandate, and Campiglio et al. (2023) quantify the "greenness" of central bank discourse. There are also interactions between embeddings (mainly from the subdomain of topic-modeling) and sentiment analysis, such as Hansen and McMahon (2016).

However, most existing studies rely on embeddings that were trained on broad, general-purpose corpora, such as Wikipedia articles. Shapiro et al. (2020), for instance, use Pennington et al.'s (2014) embeddings in their analysis of news articles. The authors are unconvinced by the results and resort to the modified dictionary approach discussed earlier. However, as we show in Section 3, the lack of predictive power may not reflect an inherent shortcoming in the use of embeddings per se. Rather, it may be indicative of the embeddings being trained on a corpus from a domain that is overly general. When we tested different publicly available language models, we found that many fail to include representations for key monetary policy-specific terms, such as HICP. Newer models such as ChatGPT even rely on undisclosed training data and are potentially prone to data leakage. As these models are primarily designed to be generative, they can be inconsistent across queries, undermining reproducibility – a fundamental requirement for rigorous empirical research.

Notable exceptions, and thus methodologically closest to our paper, are Apel et al. (2019), Bertsch et al. (2022), Araci (2019), all of which train or fine-tune embeddings for domain-specific applications. However, these studies differ in both their training corpora and modeling approaches. Apel et al. (2019) train a recurrent neural network on U.S. congressional debates to quantify disagreement among central bankers. Bertsch et al. (2022) employs a transformer-based model trained on Fed speeches to analyze the Fed's interpretation of its financial stability mandate. Araci (2019) fine-tune a transformer model on corporate financial communications

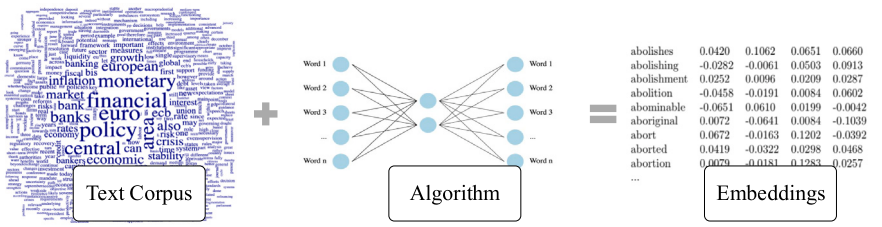


Fig. 2. How to retrieve embeddings.
Note: This graph outlines the paper's structure. In Section 3.1, we introduce the central bank communication novel text corpus. Section 3.2 details the various machine learning algorithms employed. Training these algorithms on the text corpus yields the embeddings, which we evaluate in Section 4 and subsequently apply in Section 5.

to assess analysts' sentiment. In all three cases, word embeddings are a byproduct rather than the primary focus, and the reliance on training corpora centered on parliamentary debates, Fed communications, and corporate finance limits their suitability for the broader quantification of central bank communication. Additionally, transformer architectures introduce added complexity, making it more challenging to integrate their embeddings into downstream empirical analyses. However, since these models rely on (sequences of) embeddings as inputs, they could potentially benefit from incorporating domain-specific embeddings, such as those developed in this paper, as a more efficient starting point.

To the best of our knowledge, we are the first paper to train embeddings on a large central bank communication corpus. Thereby, this paper contributes to two current strands in this literature. On the one hand, the development of novel text representations, and on the other hand, the need to fine-tune these representations for their respective use.

3. Methodology

> *"The meaning of words lies in their use. [...] One cannot guess how a word functions. One has to look at its use, and learn from that".*
>
> — Wittgenstein (1958, p. 80)

A language model maps a text corpus into an n-dimensional space, whereby the model itself can be arbitrarily simple. Take, for instance, a dictionary that classifies terms as positive, negative, and neutral, thereby mapping a corpus' vocabulary into a single dimension. This paper's proposed language model is a multidimensional representation called embedding, trained on a large domain-specific text corpus. The resulting text representation can then be applied in downstream econometric models. By separating the training process from its application, we implement *transfer learning*, a process in which specialized knowledge is gained from one task and is subsequently applied to a different, but related, task. As a result, our approach avoids potential conflicts that arise when dimension reduction and the application of dimension-reduced variables are performed simultaneously (*e.g.*, Egami et al., 2018). Fig. 2 outlines our approach, pairing a central bank-specific text corpus (Section 3.1) with a machine learning algorithm (introduced in Section 3.2 and evaluated in Section 4) to generate domain-specific embeddings that are subsequently applied in Section 5.

3.1. Text corpus

To ensure domain-specificity, our corpus focuses exclusively on central bank communication. To achieve this, we acquire all English central bank speeches published by the Bank for International Settlements (BIS), along with reports, minutes, forecasts, press conferences, and economic reviews gathered from central bank websites. Table 1 provides an overview of the sources we use for the corpus.

After removing duplicates, our corpus consists of 21,916 documents comprising 112 million words from 128 central banks. Table 2 provides a breakdown of each central bank's contribution. As expected, the Fed, Nippon Ginkō (the Bank of Japan), and the ECB are the largest contributors, collectively accounting for around 40% of the corpus. However, emerging markets are also well represented: the Reserve Bank of India ranks fourth with over 800 speeches (approximately 4% of the corpus), and the Bank Negara Malaysia (the Central Bank of Malaysia) accounts for over 450 speeches (around 2%). Fig. 3 provides further properties of our corpus. For instance, in terms of the temporal distribution, the majority of the documents stem from the past two decades. In fact, around 93% of the text stems from the post-2000 period, with the majority (55%) originating in the 2010s. In terms of geographic distribution, we cover countries representing 83% of the global population and 89% of global GDP (based on 2007 data).

We apply a minimum of pre-processing to the text corpus, in contrast to most of the central bank communication literature (*e.g.*, Hansen and McMahon, 2016; Ehrmann and Talmi, 2020). Little pre-processing is common in the embeddings literature (*e.g.*, Mikolov

Table 1
Sources for the central bank communication corpus.

Source	Type	Number of documents
BIS	Speeches	16,627
Fed	Minutes, Press Conferences, Transcripts, Agendas, Blue-, Green-, Teal-, Beige- and Red-Books	2238
BoJ	Minutes, Economic Reports, Releases, Outlook Reports	2187
ECB	Minutes, Press Conferences, Economic Outlooks, Blogs	343
Riksbank	Minutes, Economic Reviews, Monetary Policy Reports	330
Australia	Minutes	159
Poland	Minutes	156
Iceland	Minutes	101

Table 2
Central bank communication corpus summary.

	Central bank	Number of speeches	Fraction of the corpus
1	US Federal Reserve	3706	16.9%
2	Bank of Japan	2779	12.7%
3	European Central Bank	2481	11.3%
4	Reserve Bank of India	814	3.7%
5	Sveriges Riksbank	800	3.7%
6	Deutsche Bundesbank	703	3.2%
7	Bank of England	659	3.0%
8	Reserve Bank of Australia	644	2.9%
9	Bank of Canada	505	2.3%
10	Central Bank of Malaysia	467	2.1%
	...		
128	Banco de Guatemala	1	0.005%

Note: Own calculations; based on text corpus as described in Section 3.

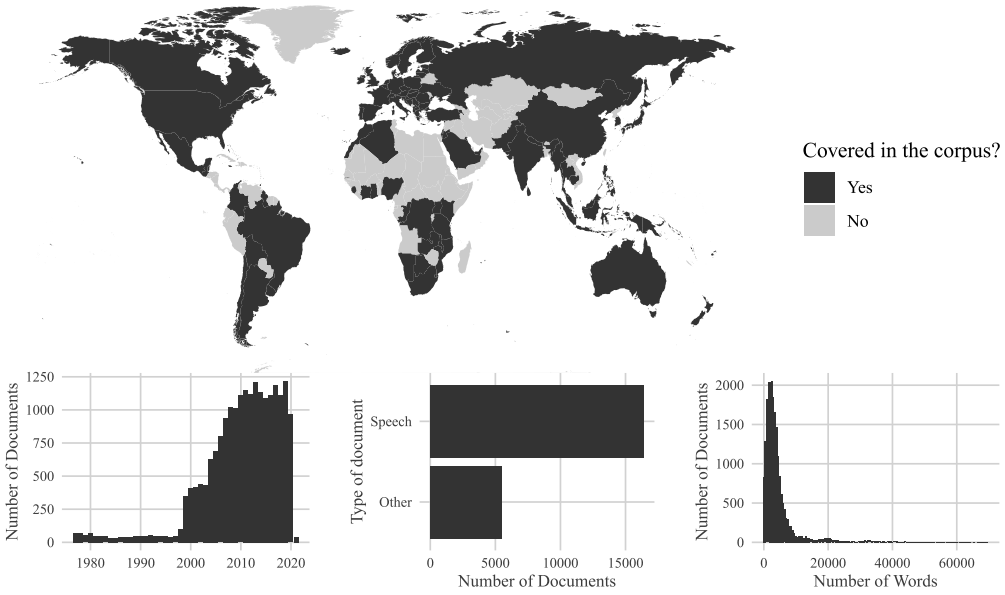


Fig. 3. Properties of the central bank communication corpus.

et al., 2013a) since similar words should be in near proximity in the vector space, which eliminates the need for procedures such as stemming, lemmatization, or stopword removal. As a result, we limit pre-processing to convert the entire text to lowercase, replacing unique entities – such as email addresses or URLs – with generic placeholder tokens, and identifying so-called collocations—word pairs with a distinct meaning when used together, such as FEDERAL and RESERVE. To efficiently identify the set of relevant collocations, we rely on Blaheta and Johnson's (2001) algorithm, which we then complement with all speaker names, thus BEN and BERNANKE become BEN_BERNANKE.

3.2. Algorithm

Modern language models largely follow Zellig Harris's (1954) distributional hypothesis, whereas the meaning of a word can be approximated by examining the distribution of the contexts in which it occurs. Specifically, if two words frequently appear in similar contexts, they likely represent similar concepts, whereas the difference in context reflects the difference in meaning. To illustrate, consider the term FORECAST as used by then-Chairman Ben Bernanke in 2005:

> *"[...] for example, the blue chip consensus* ***forecast*** *released yesterday looks for real growth 3.6 percent this year [...]*
>
> — Ben Bernanke at the Finance Committee Luncheon, 8 March 2005

The adjacent (underlined) words surrounding the so-called target word FORECAST form the word's six-word context window. According to Harris (1954), words that frequently occur in similar contexts tend to have related meanings. For instance, in a 2011 speech, then-Vice Chair Janet Yellen used the semantically similar term OUTLOOK in a comparable context:

> *"[...] for example, the blue chip consensus* ***outlook*** *for real GDP growth has edged down only modestly [...]*
>
> — Janet Yellen at the Economic Club of New York, 11 April 2011

This recurring pattern is not coincidental. In fact, the words THE, BLUE, CHIP, and CONSENSUS are followed by the word OUTLOOK 32 times and FORECAST ten times in our corpus, emphasizing their semantic similarity based on shared contexts.

Harris's (1954) distributional hypothesis is directly incorporated into modern language models during the training process, which results in a numerical vector representation for each word, known as *word embedding*. For instance, the words FORECAST and OUTLOOK will later be represented by the following vectors:

$$\mathrm{v}_{forecast} = [-0.063, -0.026, 0.0007, \ldots]$$
$$\mathrm{v}_{outlook} = [-0.051, -0.060, -0.015, \ldots]$$

These vector representations capture word similarities in a continuous, high-dimensional space, where words with similar meanings – such as FORECAST and OUTLOOK – are positioned close to each other.

This proximity is achieved by training a neural network to predict the target word, based on its surrounding context. To illustrate, the network is tasked to predict the missing word in the following sequence:

FOR EXAMPLE THE BLUE CHIP CONSENSUS _____ RELEASED YESTERDAY LOOKS FOR REAL GROWTH

In order to achieve an accurate prediction on average, the neural network has to assign high probabilities to both OUTLOOK and FORECAST. As a consequence of the prediction task, the network places these words in close proximity in the word embedding space, thereby capturing the semantic meaning as a byproduct. By contrast, a term like GDP, which is less probable in this particular context, is positioned farther away in the embedding space, reflecting its dissimilar meaning.

The above-described algorithm, called *Word2Vec* (Mikolov et al., 2013b,a,c), is a popular prediction-based model that employs neural networks with a single linear hidden layer connected to a softmax output layer.

While word embeddings capture the meaning of individual words, document embeddings extend this concept to entire documents (*e.g.*, speeches), whilst retaining the same properties. For instance, the document embeddings for the earlier speeches by Bernanke and Yellen will later be represented by the following vectors:

$$\mathrm{v}_{Bernanke\ speech} = [-0.034, 0.028, 0.043, \ldots]$$
$$\mathrm{v}_{Yellen\ speech} = [-0.038, -0.042, 0.020, \ldots]$$

Document embeddings capture the broader themes of a speech. For instance, although both Bernanke's and Yellen's speeches use similar context words (BLUE, CHIP, CONSENSUS, ...) when describing OUTLOOK and FORECAST, the speeches differ in the broader themes: Bernanke discusses real growth above 3%, while Yellen addresses an economic slowdown. Such nuances, driven by differences in the broader narrative, cannot be fully captured by the adjacent words alone. Document embeddings encode this higher-order information, serving a role analogous to including (document-specific) fixed effects in a traditional regression model to account for each document's unique characteristics.

Doc2Vec (Le and Mikolov, 2014) is an extension to Word2Vec, which trains document embeddings alongside word embeddings. An illustration of the Doc2Vec model is provided in Fig. 4.

An alternative to the prediction-based method for training embeddings described above leverages corpus-wide statistics to obtain text representations, using algorithms such as Latent Dirichlet allocation (LDA) or GloVe (*e.g.*, Blei et al., 2003; Pennington et al., 2014). We will demonstrate, however, that prediction-based methods outperform these corpus-wide approaches. A comprehensive introduction to Word2Vec, Doc2Vec, LDA, and GloVe is provided in the online appendix.

Finally, an alternative to training embeddings from scratch is the use of pre-trained models, that are freely available and are commonly trained on large, albeit general, corpora. To evaluate the quality of our embeddings, we benchmark them against two of the most widely used pre-trained word embeddings: Pennington et al.'s (2014) Glove6B and Le and Mikolov's (2014) Word2Vec GoogleNews. GloVe6B is trained on a corpus of Wikipedia and news articles, while Word2Vec GoogleNews is trained on a corpus of Google News articles. As such, both are trained on datasets considerably larger than ours, with GloVe6B utilizing 6 billion words compared to our 100 million words.

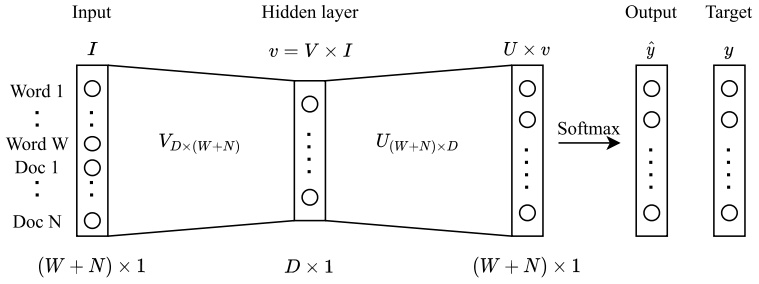


Fig. 4. Graphical illustration of Le and Mikolov (2014)'s Doc2Vec model.
Note: This illustration depicts the architecture of the Doc2Vec model, as outlined by Le and Mikolov (2014). The model projects both word embeddings and document embeddings by mapping the input layer into the hidden layer, effectively capturing semantic relationships for downstream applications.

3.3. Rhetoric stability

If, as postulated by Harris (1954), context defines meaning, then the stability of that context is a necessary condition for inference to consistently map input words to feature values. Consequently, when the context of a word differs significantly between the training and application, the resulting analysis risks producing biased estimates. We term the condition where the training and the application share a stable context *rhetorical stability*.[1] In the following, we propose a test to assess rhetorical stability between two corpora.

Our test starts by retrieving the context words (*i.e.* the surrounding terms) for key concepts from both corpora, in our case monetary policy terms such as the term INFLATION. If the context words occur with similar frequency in the training and application corpus, it can be concluded that the key concepts have the same meaning in both corpora. We evaluate rhetorical stability along two dimensions: (i) between Wikipedia articles and central bank communication and (ii) across central banks. The first dimension examines whether general-purpose corpora, such as Wikipedia articles, can effectively train models for use in central bank communication. The second dimension investigates whether communication from one central bank, such as the ECB, can effectively train models for application to another central bank, like the Fed.

We start by obtaining two representative text samples from Wikipedia to analyze the first dimension. The first sample, referred to as "Wikipedia Random", consists of 1,000 randomly selected Wikipedia articles using the *getwiki* package in *R*. The second sample, "Wikipedia Monetary Policy", is designed to evaluate the potential merits of a specialized corpus. It consists of the top twenty Wikipedia articles discussing monetary policy terms (described in the next paragraph). For the second dimension, we retrieve all speeches of the most prominent central banks in our corpus: the Fed, ECB, BoE, BoJ, Bundesbank, Riksbank, BoC, and RBI.

To identify relevant to monetary policy words, we use Apel and Grimaldi (2014)'s Monetary Policy Dictionary. Specifically, we use the following nine terms: INFLAT*, PRICE, WAGE, CYCLICAL, GROWTH, EMPLOY, UNEMPLO*, RECOVER*, and COST, with the asterisk serving as a placeholder. For each monetary policy term, we retrieve all words occurring within a six-word context and standardize the frequencies to allow comparison across datasets. For instance, the most frequent term in the context of PRICE in Wikipedia articles is SHARE (0.5%), whereas it is STABILITY for the Fed (0.2%) and the ECB (0.9%). In contrast, STABILITY as a context word for the term INFLATION is virtually non-existent in the Wikipedia corpus. To formally test rhetorical stability, we compare the relative frequency f of the context word j between the Fed and the different samples k =(Wiki, Wiki MP, ECB, BoE, …) for each target word i:

$$f_{i,j,Fed} = \beta_0 + \beta_1 f_{i,k} + \beta_2 i + \epsilon \tag{1}$$

The results of these regressions are presented in Table 3, highlighting three key findings. First, column one shows only a small coefficient for the "Wikipedia Random" corpus, indicating a weak correlation between the context words used by the Fed and Wikipedia articles. The explanatory power is also weak, with an R^2 of just 2%. Second, column two suggests that monetary policy-focused Wikipedia articles perform notably better. The correlation improves threefold and the model fits fourfold, suggesting that thematic specificity in the corpus significantly enhances context stability. Finally, the coefficients across central banks consistently exceed 0.45, a fivefold increase over the Wikipedia Random sample, with the explained variance rising from 2%–8% to 35%–45%. This significant improvement provides strong evidence that rhetorical stability is substantially greater across central bank speeches compared to Wikipedia articles.

In summary, our findings suggest that the language used by monetary policymakers differs substantially from that found in publicly available sources such as Wikipedia articles, yet remains relatively stable across central banks. This implies that the meaning of key terms, such as INFLATION, differs in Wikipedia articles compared to our corpus. Such differences may help explain the poor performance of embeddings in Shapiro et al. (2020), further reinforcing our thesis that central bank communication is best captured by specialized embeddings trained exclusively on central bank communication.

[1] An example of rhetorical instability would be the Google Flu Trends Project (Lazer et al., 2014), which used Google searches from 2003–2008 to predict flu-related medical appointments. The project was discontinued in 2015 due to severe misjudgments by the algorithm resulting from changes in search behavior.

Table 3
Rhetoric stability.

	Dependent variable:							
	'Federal Reserve'							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
'Wikipedia Random'	0.10***							
	(0.004)							
'Wikipedia Monetary Policy'		0.26***						
		(0.003)						
'European Central Bank'			0.65***					
			(0.002)					
'Bank of England'				0.71***				
				(0.002)				
'Bank of Japan'					0.42***			
					(0.002)			
'Bank of Canada'						0.46***		
						(0.002)		
'Sveriges Riksbank'							0.50***	
							(0.002)	
'Reserve Bank of India'								0.57***
								(0.002)
Keyword Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	104,267	105,109	116,645	110,231	110,314	107,280	120,128	109,352
R^2	0.02	0.09	0.45	0.45	0.33	0.33	0.39	0.34

Note: Coefficients are estimated using an OLS regression. Standard errors are displayed in parentheses. ***, **, * indicate significance at the 1, 5, and 10 per cent level, respectively. The dependent and independent variable are relative term frequencies as defined in Section 3.3.

4. Evaluation of embeddings

In this section, we train and evaluate the algorithms introduced in Section 3. Due to the algorithm's heterogeneity – Doc2Vec and LDA generate both word and document embeddings – we estimate word and document representations jointly whenever possible. Since no benchmark exists for evaluating language models in economics yet, we adopt evaluation practices from computational linguistics, where tasks can be broadly distinguished into intrinsic or extrinsic. Extrinsic evaluations test the embeddings' ability to solve specific tasks in known contexts, such as predictive modeling. Intrinsic evaluations examine whether the embeddings capture assumed relationships between words, focusing on how well the word representations align with our perception of the concepts those words represent.

We conduct two extrinsic evaluations – one focused on linguistics performance and the other on monetary policy shock predictions – to identify the best-performing embeddings. Following this, we conduct three intrinsic evaluations on the selected embeddings, comparing it against pre-trained models.

4.1. Extrinsic evaluation 1: Word predictions

The first evaluation takes advantage of Harris's (1954) distributional theory, whereas a well-trained language model should be able to predict a missing word based on its surrounding context. To illustrate this, let us revisit the example from earlier:

FOR EXAMPLE THE BLUE CHIP CONSENSUS _____ RELEASED YESTERDAY LOOKS FOR REAL GROWTH

The missing word here is FORECAST. The model prediction is considered correct only if the prediction is the word FORECAST. All embeddings are trained on this task using a neural network architecture identical to the one described in the previous chapter. The embeddings are assessed on their out-of-sample predictive performance using a held-out 10% of the corpus. Accuracy, the evaluation metric, is calculated as the number of correct predictions divided by the total number of predictions. To ensure robust and reliable evaluation estimates, we use 10-fold cross-validation, which averages results across different data splits. The results of this evaluation are presented in Table 4, where the best performance is underlined.

First, with the exception of LDA, all models achieve word prediction accuracies above 40%, thereby correctly predicting almost half of the missing words. Second, the relative performance of these models follows the order: LDA, Word2Vec, and then GloVe and Doc2Vec, with Doc2Vec (word prediction-based) and GloVe (corpus-wide statistics) performing similarly, highlighting the benefits of a diverse approach. Finally, Doc2Vec achieves the highest accuracy, correctly predicting 82% of terms, which represents a 20–30 percentage-point improvement over the publicly available models (Word2Vec GoogleNews and GloVe6B).

Table 4
Extrinsic evaluation 1: Word prediction.

Algorithm	Accuracy	Standard deviation
LDA	0.05	0.008
Word2Vec GoogleNews	0.47	0.012
Word2Vec Bow	0.57	0.006
GloVe6B	0.63	0.008
Word2Vec Word2Vec Skipgram	0.66	0.004
Doc2Vec PVDM Pre	0.74	0.029
Doc2Vec PVDM	0.75	0.020
GloVe	0.81	0.007
Doc2Vec Bow	0.82	0.011
Doc2Vec Bow Pre	0.82	0.008

Note: Accuracy is calculated as the number of correct predictions divided by the total number of predictions. Standard deviation is calculated using 10-fold cross-validation. Specifications: Bow = (Distributed) Bag Of Words; PVDM = Paragraph Vector Distributed Memory; Pre = pre-trained embeddings were used as more efficient starting points.

4.2. Extrinsic evaluation 2: Monetary policy shock predictions

Our second extrinsic evaluation task addresses a standard empirical exercise in the field of central bank communication: improving a traditional Taylor rule by incorporating textual information. Following Aruoba and Drechsel (2022), we isolate the explanatory power of central bank communication beyond traditional macroeconomic variables.

We begin by estimating a traditional Taylor rule, on a monthly basis, for both the Fed and the ECB. Specifically, we regress the short-term interest rate (Wu and Xia (2016)-shadow rate) on a set of macroeconomic variables, namely the inflation rate, GDP growth, and the current unemployment rate. The residual (ϵ) captures the unexplained variation in the interest rate after accounting for the included macroeconomic variables. In other words, it reflects a standard monetary policy shock.

We then test which language model best explains the monetary policy shock. For each speech i delivered at time t, we either (i) retrieve the corresponding document embedding directly or (ii) calculate the document representation from the word embeddings by computing the dot product of the word embedding matrix with the term frequency matrix for the corresponding speech. In order to compare performance with dictionary-based approaches, we (iii) incorporate four established dictionaries from monetary economics (Bennani and Neuenkirch, 2017), finance (Loughran and McDonald, 2011), and computational linguistics (Hu and Liu, 2004; Mohammad and Turney, 2013). To provide the dictionaries with the optimal conditions, we integrate both the absolute frequency of the terms they identify (*e.g.*, the number of positive terms) and, if applicable, relative measures such as a sentiment score or a hawkish-dovish index. Acknowledging the potential complementarity of dictionaries (*e.g.*, Shapiro et al., 2020), we also assess a specification that combines all four dictionaries.

To address the high dimensionality of speech representations (up to 300 dimensions) – which may result in over-fitting and limit generalizability in a standard OLS regression – we apply Elastic Net regularization (*e.g.*, Zou and Hastie, 2005), as used in Aruoba and Drechsel (2022). Elastic Nets identify and remove potentially irrelevant dimensions. To assess the marginal explanatory power of textual information, we thus run the following regression:

$$\epsilon_{i,t} = h(d_{i,t}) + \nu_{i,t}, \tag{2}$$

where $\epsilon_{i,t}$ is the monetary policy shock, $d_{i,t}$ the speech representation, $h(.)$ the nonlinear regularization function of the Elastic Net, and $\nu_{i,t}$ the error term. Each model is trained on 80% of the observations and its performance is evaluated out-of-sample using the remaining 20%. The results, presented in Table 5, show the relative improvement of each approach in explaining the monetary policy shock (*i.e.* ν/ϵ), with lower values indicating better performance.

Table 5 shows that while dictionary-based approaches capture information, their overall impact is limited, with, for instance, Loughran and McDonald's (2011) sentiment dictionary explaining around 10% of the Fed's monetary policy shock. At best, when combining all dictionaries (resulting in a 24-dimensional representation), they reduce the unexplained variance by 15% (Fed) and 30% (ECB), which is around 10 percentage points less than the reduction achieved using the pre-trained word-embedding model by Le and Mikolov (2014).

Consistent with the findings from Extrinsic Evaluation 1, the Doc2Vec models outperform the other models at both the word-embedding and the document-embedding level. At the word level, Doc2Vec models account for approximately 30% (Fed) to 50% (ECB) of the monetary policy shock. At the document level, their explanatory power increases to 50% (Fed) and 70% (ECB). In light of its superior performance in both word prediction and interest rate prediction, we choose the "bag-of-words with pre-training" specification of the Doc2Vec models (bolded in all tables) as our primary embedding for the remainder of the paper. For the sake of brevity, we will refer to this model as "Doc2Vec" henceforth.

4.3. Intrinsic evaluation 1: Macroeconomic concepts

The first intrinsic evaluation assesses whether the selected Doc2Vec model effectively positions key macroeconomic terms (INFLATION, UNEMPLOYMENT, and OUTPUT) close to words with similar meanings in the vector space. To identify the closest terms for

Table 5
Extrinsic evaluation 2: Monetary policy shock predictions.

	Monetary policy shocks	
	Federal Reserve	European Central Bank
No model	1.00	1.00
Dictionaries		
Loughran-McDonald Sentiment	0.89	0.82
Hu Lui Sentiment	0.88	0.78
Hawk-Dove Dictionary	0.88	0.81
NRC Word-Emotion Lexicon	0.87	0.79
All Dictionaries combined	0.85	0.70
Word Embeddings		
Word2Vec GoogleNews	0.75	0.61
GloVe6B	0.74	0.60
LDA	0.75	0.62
Word2Vec Bow	0.75	0.58
GloVe	0.72	0.56
Word2Vec Skipgram	0.70	0.58
Doc2Vec Bow	0.70	0.56
Doc2Vec Bow Pre	0.74	0.54
Doc2Vec PVDM	0.73	0.53
Doc2Vec PVDM Pre	0.68	0.52
Document Embeddings		
LDA	0.56	0.42
Doc2Vec PVDM	0.56	0.37
Doc2Vec PVDM Pre	0.56	0.37
Doc2Vec Bow	0.50	0.33
Doc2Vec Bow Pre	0.55	0.30

Note: Bow = (Distributed) Bag Of Words; PVDM = Paragraph Vector Distributed Memory; Pre = pre-trained embeddings were used as more efficient starting points. The dictionaries are based on Loughran and McDonald (2011), Hu and Liu (2004), Bennani and Neuenkirch (2017), Mohammad and Turney (2013). We include both, the absolute number of identified terms, as well as the relative number (for instance the *sentiment*).

each of these macroeconomic concepts, we rely on cosine similarity (S), which calculates the proximity of two vectors a and b as follows:

$$S_{a,b} = \frac{a \cdot b}{\|a\| \times \|b\|} \tag{3}$$

Consequently, words with high similarity are expected to be semantically close due to their relative positioning in the embedding space. Table 6 presents the ten most similar terms in the vocabulary for each of the macroeconomic concepts. The results suggest that the Doc2Vec model effectively groups words with similar economic meanings. For instance, the model associates terms such as CORE_INFLATION and INFLATION_EXPECTATIONS with INFLATION. Additionally, the Doc2Vec model captures broader relationships between economic concepts, as evidenced by the high similarity between UNEMPLOYMENT and ECONOMIC_SLACK.[2]

4.4. Intrinsic evaluation 2: Homonyms

The second intrinsic evaluation addresses a significant challenge for all language models: homonyms—words with multiple meanings depending on context. For instance, the term LIABILITY carries distinct meaning depending on the domain as discussed earlier (Loughran and McDonald, 2011). Similarly, the term BASEL may refer to either the Swiss city or the Basel Accords. Since each word occupies a single point in a high-dimensional embedding space, homonyms can introduce noise into those models. We therefore repeat the similarity exercise from the first intrinsic evaluation with the term BASEL, comparing the results of our model with those of the publicly available embeddings (GloVe6B and GoogleNews). These general embeddings are expected to primarily associate BASEL with the city, thereby limiting their reliability in quantifying central bank communication.

Table 7 presents the results, where, as expected, the latter two embeddings primarily associate BASEL with the city. In contrast, our Doc2Vec embeddings associate BASEL with banking regulation, correctly identifying related terms such as Basel Committee on Banking Supervision (BCBS). While the association of BASEL with the city may be suitable in general-purpose contexts, for monetary policy communication, the regulatory interpretation of the term is clearly more relevant, underscoring the advantage of employing domain-specific embeddings.

[2] The accompanying website to this paper (https://sites.google.com/view/whatever-it-takes-bz2021) contains an interactive tool that enables users to assess the most similar terms for any word in the entire vocabulary.

Table 6
Intrinsic evaluation 1: Similarity to key monetary policy terms.

Inflation	Unemployment	Output
core_inflation	unemployment_rate	nonfarm_business
inflation_expectations	natural_rate	sector
economic_slack	joblessness	per_hour
underlying_inflation	jobless	output_growth
inflation_outlook	labor_force	producers
price_inflation	unemployed	manufacturing_output
actual_inflation	labor_market	factory
disinflationary	economic_slack	hourly_compensation
inflation_rate	unemployment_rates	business_equipment
disinflation	participation_rate	labor_costs

Note: The table shows the most similar terms to the words INFLATION, UNEMPLOYMENT, and OUTPUT according to the cosine distance of the underlying word embeddings of the Doc2Vec model chosen in Section 4.2. The underscore highlights collocations as described in Section 3.

Table 7
Intrinsic evaluation 2: Similarity to BASEL across language models.

Doc2Vec	GloVe6B	Word2Vec GoogleNews
basel_committee	zurich	abbr
basle	basle	Tst
capital_accord	zürich	iva
basel_accord	bern	tHe
bcbs	switzerland	Neurol
basle_committee	stuttgart	BASLE
basel_ii	hamburg	PARAGRAPH
basel_iii	cologne	tellus
consultative	lausanne	Def.
minimum_capital	schaffhausen	Complementarity

Note: Ten most similar words to the word BASEL according to the cosine distance of the underlying word embeddings as defined by Eq. (3). The underscore highlights collocations as described in Section 3.

4.5. Intrinsic evaluation 3: Central bank similarities

Finally, we conduct an intrinsic evaluation of the document embeddings. We posit that central banks in Western economies, given their similar mandates and economic structures, communicate more similarly compared to central banks in other regions. Fig. 5 illustrates the similarity of each central bank to the ECB via a comparison of their average document embeddings. Darker colors represent higher similarity. The results lend support to our hypothesis. Central banks in Europe and North America appear to be most similar to the ECB, which serves as a starting point for our subsequent analysis of monetary policy regimes in the next chapter.

In summary, this section evaluated the presented algorithms, selecting Doc2Vec for its superior out-of-sample performance in capturing linguistic patterns and explaining monetary policy shocks. In three intrinsic evaluations, we provide evidence that our Doc2Vec embeddings effectively capture meaningful relationships, particularly in monetary policy contexts where general language models fall short.

5. Monetary policy regime classification

To date, our embeddings have been applied in various forms to retrieve latent messages, demonstrating their versatility. These applications include the creation of an indicator measuring the ECB's commitment to act as a lender of last resort (Baumgärtner and Zahner, 2024), or measuring the scientization process of the Bank of England (BoE) (Goutsmedt et al., 2023). In this section, we introduce another application: the development of a leading indicator for the strength of the Fed's inflation targeting regime.

According to Mishkin (1999, p. 591), a key component of inflation targeting is an *"increased transparency of the monetary policy strategy through communication with the public and the markets about the plans and objectives"*. However, traditional literature identifies monetary policy regime changes retrospectively by detecting structural breaks in macroeconomic data (*e.g.*, Bae et al., 2012; Benati and Goodhart, 2010; Bikbov and Chernov, 2013), making predictions difficult (*e.g.*, Sims and Zha, 2006) and gradual regime changes that evolve continuously difficult to measure. To address these limitations, we propose a method that identifies regime changes of the Fed based on its inter-meeting communication. Our approach emphasizes observable intention, rather than action, making it forward-looking and predictive given the accessibility of modern central bank communication.

We proceed as follows: First, we show that central bank communication differs systematically across monetary policy regimes. We then exploit these differences to construct a novel communication-based index that effectively tracks shifts in the Fed's inflation regime. This continuous, leading index tracks regimes identified by traditional methods while predicting adjustments in inflation and interest rate expectations and deviations from rule-based policy actions.

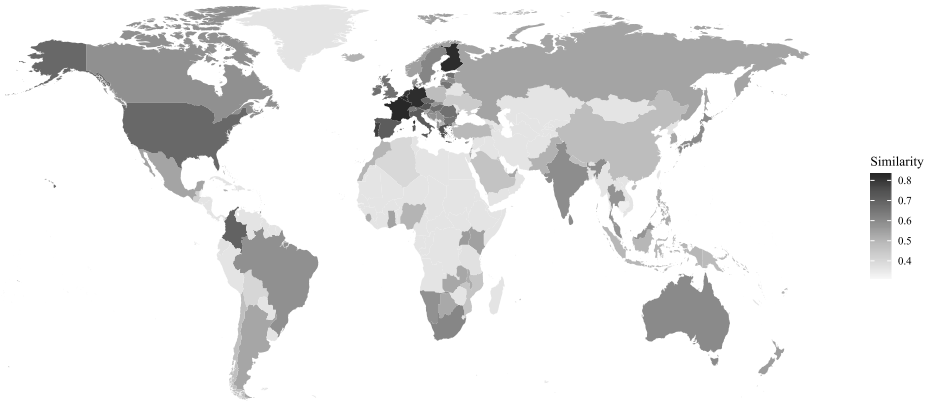


Fig. 5. Intrinsic Evaluation 3: Similarity to the ECB.
Note: This graph illustrates the cosine distance (defined in Eq. (3)) between the average ECB document embedding and all average central bank document embeddings in our dataset. Darker colors depict a lower distance, *i.e.* a higher similarity.

5.1. Communication differences and monetary policy regimes

We begin by examining the empirical relationship between central bank communication and monetary policy regimes. To conduct this analysis, we combine two datasets. The first captures communication similarities between central banks, and the second classifies their respective monetary policy regime.

The first dataset measures the annual communication similarity of a central bank and a reference group (the Reserve Bank of New Zealand (RBNZ), the Fed, and the ECB). Specifically, using our Doc2Vec document embeddings, we compute the annual cosine distance between each central bank and the reference banks, providing the panel equivalent to the cross-sectional approach of the last intrinsic evaluation (illustrated in Fig. 5). The distances indicate the extent to which a bank's communication aligns with or diverges from the reference bank's communication on an annual basis.

The second dataset comprises the annual monetary policy regime classification by Cobham (2021).[3] Cobham's classification captures the diversity of policy regimes through ten key target variables (*e.g.*, inflation rate, money supply, …), that are further subdivided into 32 mutually distinct regimes. We focus on two of that regimes in particular:

- FIT (Full Inflation Targeting): A regime characterized by a well-defined, consistently enforced numerical inflation target and associated with transparency and predictability of monetary policy.
- LIT (Loose Inflation Targeting): A regime characterized by a flexible stance on its inflation target, allowing the central bank to respond to a broader set of economic indicators and conditions.

Among reference banks, the RBNZ was a FIT throughout the sample period, the ECB adhered to LIT for the entire period, while the Fed has transitioned from the LIT regime to the FIT regime.

Merging both datasets results in 957 bank-year observations for 88 individual central banks, which we complement by the national inflation rate, unemployment rate, and log(GDP) as controls. Table A.1 provides the respective descriptive statistics.

The empirical identification strategy is straightforward. We regress the communication similarity ($S_{i,j}$) of bank j and reference central bank i in year t on dummy variables representing the respective monetary policy regime of bank j ($Regime_j$). We account for relative macroeconomic conditions ($X_i - X_j$) and time-specific variation using year-fixed effects:

$$S_{i,j,t} = \beta_1 Regime_{j,t} + \beta_2 (X_i - X_j) + \epsilon_{i,j,t} \tag{4}$$

The coefficients of interest are the β_1's, specifically those on inflation targeting regimes, which we report in Table 8.

We find the following results. A comparison of means between inflation targeting central banks and those with other monetary policy objectives (column 1) shows a positive and significant coefficient for the RBNZ, Fed and ECB regressions. The coefficient suggests that being classified as an inflation targeting central bank increases the similarity to the reference banks from 34%–40% to 45%–52%, or by more than one standard deviation (see Table A.1). Importantly, the result persists after controlling for macroeconomic conditions (column 2).

[3] We use monetary policy frameworks and regimes interchangeably throughout the paper. While frameworks are defined as the *"objectives pursued by the monetary authorities, but also the set of constraints and conventions within which their monetary policy decisions are taken"* (Cobham, 2021, p. 1), a monetary policy regime is typically defined more narrowly focusing on monetary policy objective, target, and instrument (*e.g.*, Bae et al., 2012). However, since a monetary policy regime is inherently embedded within a broader policy framework, any change in the framework necessarily entails a change in the regime. The monetary policy classifications are available at https://monetaryframeworks.org. For members of a currency union, we assign the classification of the union's central bank.

Table 8
Regression results: Monetary policy regime classification.

	Dependent variable: Similarity towards i								
i =	RBNZ			Fed			ECB		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Inflation Target	0.11***	0.09***		0.12***	0.09***		0.15***	0.15***	
	(0.02)	(0.02)		(0.02)	(0.02)		(0.02)	(0.03)	
– FIT			0.14***			0.12***			0.10***
			(0.02)			(0.02)			(0.02)
– LIT			0.09***			0.10***			0.17***
			(0.02)			(0.02)			(0.02)
Constant	0.34***	0.35***	0.34***	0.40***	0.42***	0.41***	0.37***	0.36***	0.37***
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Rem. MPF Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Macro-controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	957	957	957	957	957	957	957	957	957
R^2	0.22	0.24	0.28	0.18	0.19	0.22	0.26	0.32	0.38
Adjusted R^2	0.20	0.21	0.26	0.16	0.17	0.19	0.24	0.30	0.36

Note: Coefficients are estimated using an OLS regression. Standard errors are displayed in parentheses. ***, **, * indicate significance at the 1, 5, and 10 per cent level, respectively. We adapt the notations directly from Cobham (2021): LIT = loose inflation targeting; FIT = full inflation targeting. *Rem. MPF Controls* indicates controls for all monetary policy frameworks not shown in the table. The full regression table can be found in the Online Appendix.

Next, we leverage the full range of classifications from Cobham's (2021) in column 3 by further subdividing inflation targeting regimes into FIT and LIT. The results are interesting both within and across the three targeting central banks. For the RBNZ and the ECB, which have consistently been classified as a FIT and LIT respectively, we find that communication similarity is largely driven by banks that share this classification. In contrast, for the Fed, which transitioned from LIT to FIT in 2011, communication similarity is evenly distributed between the two regimes. These findings make us confident that one of the key factors driving similarity in central bank communication is the adoption of a common monetary policy regime.

5.2. The Fed's inflation targeting regime

To measure how closely the Fed's communication aligns with that of LIT and FIT regimes over time, we develop an "inflation targeting" (IT) index, which quantifies the relative textual similarity of each Fed speech to both regimes. Specifically, we compute the Euclidean distance between a given speech and (i) the average FIT speech, as well as (ii) the average FIT speech in our corpus. The IT index is then obtained by the difference these two–a measure proposed by Garg et al. (2018) for comparisons with categorical classifications:

$$IT_i = \sqrt{\left(s_i^{Fed} - \frac{1}{N}\sum_{n=1}^{N}(s_n^{FIT})\right)^2} - \sqrt{\left(s_i^{Fed} - \frac{1}{K}\sum_{k=1}^{K}(s_k^{LIT})\right)^2}. \tag{5}$$

Positive IT values indicate a tilt towards FIT regimes (stronger inflation targeting), while negative values indicate alignment with LIT regimes (greater policy flexibility). To facilitate interpretation, we de-mean and standardize the IT index, so that $IT = 1$ corresponds to one standard deviation above the mean towards the FIT regime.

Fig. 6 presents the average quarterly IT. Notably, we observe pronounced negative deviations in IT that coincide with key policy interventions where secondary objectives, such as financial stability, took precedence. For instance strong declines occur during the Global Financial Crisis when the Fed reached the zero lower bound (December 2008) and introduced and expanded QE1 (November 2008 and March 2009). A similar pattern is evident during the COVID-19 pandemic following the enactment of the CARES Act.

Fluctuations in our IT index closely track Fed regime shifts in other studies, such as the Markov switching approach of Bae et al. (2012) and the term-structure model of Bikbov and Chernov (2013). For instance, Bae et al. (2012, Figure 5) document a heightened inflation response – indicative of a FIT regime – during the Greenspan era, which closely mirrors the heightened IT index until the end of 2006. Likewise, Bikbov and Chernov (2013, Figure 1) identify shifts from active to passive monetary policy regimes after 2006, mirroring a shift towards a LIT regime in our index. These parallels strengthen our confidence that the Fed's communication patterns systematically reflect changes in their policy stance.

5.3. Full and loose inflation targeting communication patterns

Before evaluating the impact of changes in our IT index, we assess whether speeches with high IT values emphasize topics consistent with FIT regimes, such as explicit mentions of inflation targets, and vice versa.

A clear example of a FIT speech is Donald Kohn's (2003) address at the Conference on Finance and Macroeconomics in February 2003. The speech starts by emphasizing the Fed's firm commitment to an inflation target:

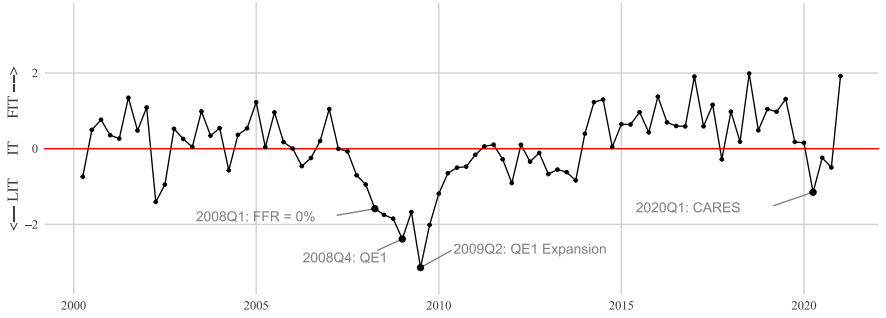


Fig. 6. Fed's stance on inflation targeting.
Note: Standardized time series of the IT index, as defined in Eq. (5). Positive values indicate closer alignment with the FIT regime, while negative values correspond to greater alignment with LIT regimes as described in Section 5.

> *"Economic contractions have frequently been led by weakness in the household sector, which often has responded to higher interest rates as the Federal Reserve acts to reverse inflation pressures"*

Kohn provides a transparent outlook on the Fed's assessment of the economy and its strategy to achieve the inflation target:

> *"With production currently well below potential and inflation expectations low, it is doubtful that the temporary misalignment of rates would result in the development of any perceptible inflation pressures before the Federal Reserve would have time to take countervailing steps.*

The speech also emphasizes the importance of policy guidance and communication:

> *"Among other things, markets could get it wrong–for example, they could anticipate greater strength in underlying demand than is actually occurring. [...] We would set rates lower than the markets have built in, and in our various statements we would attempt to make clear our assessment of economic prospects".*

By addressing a strong commitment to inflation targeting, policy guidance, and transparent communication, Kohn (2003) sets out the key characteristics of a FIT regime. As might be expected, this speech has an IT index value of over three, being one of the most FIT-oriented speeches in our dataset.

Contrast this with a clear FIT speech, such as William Dudley's (2013) remarks at the Transatlantic Economic Interdependence and Policy Challenges Conference in April 2013. The speech lacked any references to inflation, instead prioritizing discussions on financial stability targets and fiscal policy instruments:

> *"On the regulatory side, there is considerable good news worth highlighting. In particular, substantial progress has been made in strengthening the global capital and liquidity standards for internationally active banks. [...] Nevertheless, the United States could be doing better. The U.S. fiscal policy program, for example, does not appear well-calibrated to the current set of economic circumstances".*

The speech provides only vague guidance on future policy actions, primarily focusing on a retrospective account of the ongoing unconventional policy measures:

> *"To provide the appropriate degree of accommodation, the Federal Reserve has recently moved to an outcome-based approach in which the use of our tools is explicitly tied to developments in the economy and economic outlook. Currently, as part of this strategy, we are purchasing $85 billion of longer-term Treasuries and agency mortgage-backed securities each month".*

With its emphasis on flexibility and broader economic objectives, Dudley (2013) is consistent with key aspects of the LIT regime. With an index value of about -3, it ranks among the most LIT-oriented speeches in our dataset.

Both speeches further illustrate the shortcomings of traditional hawkish/dovish or sentiment-based measures. Kohn's speech employed negative and dovish language (using terms like WEAKNESS, *e.g.*, Loughran and McDonald, 2011), however, interest rates showed minimal movement in the subsequent 12 months. Conversely, William Dudley's speech was followed by substantial expansionary monetary policy actions (around 150 basis points) but would be classified as a neutral speech by both dictionaries. In contrast, our IT index provides a more accurate representation of the underlying policy regime, with a focus on inflation control in 2003 (as evidenced by the inflation rate and the interest rate moving together) and a flexible approach in 2013 (reflected by the inflation rate and the interest rate moving in opposite directions).

To expand on these anecdotal distinctions, we systematically quantify thematic differences between FIT and LIT regimes using a machine learning topic model (LDA, as also employed in Section 4). Fig. 7 presents a word cloud of the most salient topics associated with each regime.



Fig. 7. FIT/LIT Wordcloud.
Note: The word cloud illustrates the top words for a selected number of topics from the topic-word distributions of an LDA model based on Fed speeches. Black (gray) terms are primarily associated with FIT (LIT) speeches. See the Online Appendix for more details.

Black terms are predominantly associated with FIT regimes, while gray terms are linked primarily to LIT regimes. The thematic contrast is evident: FIT regime speeches tend to focus on objectives and targets, described by terms such as INFLATION, TARGET, UNEMPLOYMENT, and EXPECTATIONS, consistent with a rule-based approach to inflation targeting. In contrast, LIT regime speeches cover a broader policy scope, frequently using words related to banking supervision such as BANKING, SUPERVISORS, CREDIT, and RISK. This distinction reinforces our confidence that the IT index captures communication patterns that characterize different intensities of inflation targeting regimes.

5.4. Inflation targeting communication and inflation expectations

As Kohn (2003) highlights above, the Fed deliberately uses communication to address deviations in (market) expectations. We therefore test whether variation in communication predicts corresponding shifts in expectations, guided by two hypotheses. First, a higher IT index, signaling a stronger commitment to inflation targeting, should anchor expectations closer to the 2% target. Second, a higher IT index should also heighten the responsiveness of the Fed's policy instrument – the real interest rate – to inflation deviations. Therefore, interest rate expectations should be more sensitive to changes in inflation expectations in FIT regimes. Supporting our hypotheses, Sims and Zha (2006) and Bikbov and Chernov (2013) demonstrate that regimes with stronger policy responses to inflation deviations ("active" in their terminology, or FIT in ours) tend to be associated with greater macroeconomic stability. Accordingly, we expect higher IT to be associated with expectations closer to the target (hypothesis 1) and stronger expected policy responses to deviation from the target (hypothesis 2).

To test the first hypothesis, we regress the monthly absolute deviation of inflation expectation from the 2% target on the lagged IT index:

$$|E_t[\pi_{t+n}] - 2\%| = \alpha + \beta_1 IT_{t-1} + \epsilon_t. \tag{6}$$

We expect $\beta_1 < 0$, indicating that inflation targeting communication reduces deviations of inflation expectations from target. To control for variations in communication unrelated to IT, we control for the monthly average of uncertain language (Baker et al., 2016), sentiment (Loughran and McDonald, 2011), hawkish language (Bennani and Neuenkirch, 2017), forward guidance (Swanson, 2021), inflation risk premium, and real risk premium. Columns 1–4 of Table 9 presents the results. A one standard deviation increase in IT reduces the deviation in financial market 1-year-ahead inflation expectations by 5 basis points–one-fifth of a standard deviation (Table A.1). Similar results hold for Michigan Consumer Survey inflation expectations (column 2). The impact of IT diminishes at longer time horizons (columns 3–4).

To test whether IT moderates the expected sensitivity of monthly interest rates to inflation expectations, we adopt Bae et al.'s (2012) empirical framework:

$$E_t[r_{t+n}] = \alpha + \beta_1(E_t[\pi_{t+n}]) + \beta_2 IT_{t-1} + \beta_3 \left(E_t[\pi_{t+n}]\right) \times IT_{t-1} + \epsilon_t, \tag{7}$$

where $\beta_3 > 0$ indicates that IT strengthens the responsiveness of real interest rate expectations ($E_t[r]$) on inflation expectations ($E_t[\pi]$). Columns 5–7 of Table 9 show that the interaction terms (β_3) are both positive and significant across all horizons, mirroring the "active" regime findings of Bae et al. (2012). Their estimates place the inflation response parameter to the real interest rate between 2.5 and −0.5 which corresponds to regimes spanning from $IT = 2$ and $IT = -1$ in our framework. Again, the impact of IT diminishes at longer time horizons.

In summary, our findings emphasize the critical role of inter-meeting communication in anchoring inflation expectations and enhancing the sensitivity of interest rate expectations to inflation expectations.

Table 9
Regression results: Expectations.

	Dependent variable:						
	$\Delta E_t[\pi_{1y}]$	$\Delta E_t[\pi_{1y}^{Mich}]$	$\Delta E_t[\pi_{2y}]$	$\Delta E_t[\pi_{10y}]$	$E_t[r_{1m}]$	$E_t[r_{1y}]$	$E_t[r_{10y}]$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
IT_{t-1}	−0.05**	−0.09**	−0.02	−0.002	−0.03	0.004	−0.04
	(0.03)	(0.03)	(0.02)	(0.02)	(0.12)	(0.08)	(0.03)
$E_t[\pi_{1y}]$					0.43**	0.84***	0.95***
					(0.20)	(0.13)	(0.06)
$E_t[\pi_{1y}] \times IT_{t-1}$					0.97***	0.57***	0.29***
					(0.19)	(0.12)	(0.06)
Observations	240	240	240	240	240	240	204
R^2	0.07	0.39	0.05	0.11	0.27	0.48	0.63
Adjusted R^2	0.04	0.37	0.01	0.08	0.24	0.46	0.61

Note: Coefficients are estimated using an OLS regression. Standard errors are displayed in parentheses. ***, **, * indicate significance at the 1, 5, and 10 per cent level, respectively. Inflation expectation is measured in absolute deviation from 2%. IT is defined in Eq. (5) and measured in standard-deviations from its historical mean. All regressions include controls for forward guidance shocks (Swanson, 2021), uncertain language (Baker et al., 2016), sentiment (Loughran and McDonald, 2011), hawkish language (Bennani and Neuenkirch, 2017), inflation risk premium, and real risk premium. The full regression table can be found in the Online Appendix.

Table 10
Regression results: IT Taylor rule.

	Dependent variable:						
	Shadow Interest Rate						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$(\pi - 2\%)$	1.07***	1.33***	0.84***	0.64***	1.03***	0.84***	1.09***
	(0.20)	(0.22)	(0.16)	(0.17)	(0.17)	(0.17)	(0.18)
IT_{t-1}		0.13	0.40	0.66	−0.40	0.51	1.15
		(0.23)	(0.76)	(0.84)	(0.84)	(0.79)	(1.01)
$IT_{t-1} \times (\pi - 2\%)$		0.46**	0.81***	0.69***	0.64***	0.82***	0.71***
		(0.19)	(0.17)	(0.18)	(0.20)	(0.18)	(0.18)
Unemp. Rate			−1.02***	−1.26***	−0.80***	−1.01***	−1.02***
			(0.12)	(0.14)	(0.13)	(0.14)	(0.19)
Output Gap			0.37*	0.37*	0.30	0.44**	0.16
			(0.19)	(0.20)	(0.22)	(0.21)	(0.22)
$IT_{t-1} \times$ *Unemp. Rate*			−0.22*	−0.26*	−0.01	−0.24*	−0.29
			(0.13)	(0.14)	(0.14)	(0.14)	(0.18)
$IT_{t-1} \times$ *Output Gap*			−0.41	−0.34	0.08	−0.40	−0.12
			(0.26)	(0.28)	(0.25)	(0.28)	(0.23)
Constant	1.24***	1.10***	6.97***	8.11***	5.66***	6.22***	6.63***
	(0.23)	(0.23)	(0.74)	(0.81)	(0.79)	(1.31)	(1.04)
Observations	85	84	84	84	84	81	84
R^2	0.25	0.32	0.74	0.74	0.66	0.76	0.70
Adjusted R^2	0.24	0.29	0.72	0.71	0.63	0.72	0.68

Note: Coefficients are estimated using an OLS regression. Standard errors are displayed in parentheses. ***, **, * indicate significance at the 1, 5, and 10 per cent level, respectively. Shadow Interest Rate is from Wu and Xia (2016). IT is defined in Eq. (5) and measured in standard-deviations from its historical mean. π constitutes the annual CPI inflation rate. Output Gap is measured as the Cyclical Component of HP Filtered Real GDP Series. Column (4) uses Krippner's (2020) Short term shadow rate, column (5) controls for speaker fixed effects, column (6) controls for dictionaries (Loughran and McDonald, 2011; Bennani and Neuenkirch, 2017; Baker et al., 2016) and forward guidance shocks (Swanson, 2021), and column (7) calculates IT using an expanding rolling window approach. The full regression table can be found in the Online Appendix.

5.5. Inflation targeting communication and the Taylor rule

Building on the previous findings, we test whether changes in inflation regimes serve as leading indicators for deviations from rule-based monetary policy. Our hypothesis builds on the previous finding, suggesting that the IT index moderates the inflation response parameter in the Fed's rule-based monetary policy. The empirical framework to assess our hypothesis closely follows the previous specification. Specifically, we estimate the following augmented Taylor rule:

$$i_t = \alpha + \beta_1(\pi_t - 2\%) + \beta_2 IT_t + \beta_3 \left(\pi_t - 2\%\right) \times IT_t + \epsilon_t \tag{8}$$

where i represents the Wu and Xia (2016) shadow rate, π the current inflation rate, and IT the lagged inflation regime index. We expect $\beta_3 > 0$, implying that an increase in IT (closer to FIT regimes) leads to a stronger rule-based response to inflation deviations.

Table 10 shows, that once IT is included (columns 2–4), the inflation response coefficient depends on the inflation regime. At the historical mean ($IT \approx 0$), the inflation response parameter is about 1.3–consistent with Bae et al. (2012), who report an estimate

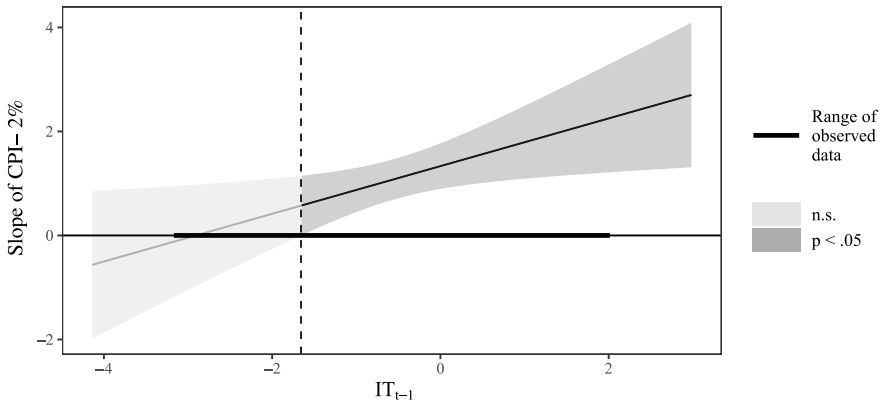


Fig. 8. Interaction effect of inflation response and IT.
Note: The graph is based on the second regression in Table 10. The Johnson-Neyman confidence interval are at the 95% significance level. The distribution of uncertainty is shown by the thin black line.

of 1.32 for 1997–2005, a period when the IT index was close to zero. A one standard deviation increase in the IT index raises the inflation response by 0.45 and vice versa, suggesting a potential Taylor principle violation. During the financial crisis ($IT \approx -2.9$) and the early COVID-19 period ($IT \approx -2.2$), the inflation response parameter fell to near zero, highlighting the discretionary stance during these periods. Fig. 8 plots the dynamic relationship between IT and the inflation response alongside Johnson-Neyman 95% level confidence, showing that high IT levels are associated with a strong inflation response, while low levels render the response insignificant. This variance in the inflation parameter across regimes is consistent with the previously cited Bae et al. (2012), and Benati and Goodhart (2010, Figure 14), who find estimates ranging from near zero to over three.

We conduct a series of robustness checks. Column 3 of Table 10 shows that our findings are robust to the inclusion of additional business cycle indicators. Furthermore, our findings are robust to the use of Krippner's (2020) shadow short rate (column 4), controlling for speaker-level heterogeneity as suggested by Hayo and Zahner (2023) (column 5), controlling for dictionary-based measures and forward guidance shocks (column 6) and, re-estimating IT with an expanding window to avoid data leakage.

To summarize, our results suggest that shifts in our communication-based Fed inflation regime index strengthen the inflation response coefficient, indicating a closer adherence to a rule-based policy during such periods. These findings reinforce the role of central bank communication in shaping market expectations and guiding subsequent policy decisions.

6. Conclusion

Understanding the communication of central banks has developed to be a substantial entity in monetary policy, with dictionary approaches being the most frequently used to quantify their press conferences, speeches, and reports. In this paper, we push the research frontier in four ways to generate high-quality text representations for central bank communication: We compile a novel text-corpus, introduce algorithms stemming from computational linguistics to extract embeddings – meaningful, multidimensional text representations –, train central bank-specific embeddings and develop a communication-based inflation-targeting regime indicator for the Federal Reserve (Fed) based on our embeddings.

First, we collect a text corpus that is unparalleled in size and diversity, with over 20,000 speeches from over 120 central banks. We show that our corpus offers significant advantages over conventional corpora used in the existing literature for quantifying central bank communication. Second, we introduce a class of machine learning algorithms from computational linguistics to generate embeddings—multidimensional vector representations for words and documents. Third, we train these algorithms on our central bank communication corpus and evaluate the resulting embeddings across multiple tasks. Our results show that they outperform existing embeddings and conventional dictionaries in predicting monetary policy shocks. We provide evidence that the superior performance stems from the domain specific corpus. Finally, we use the best-performing embeddings to develop a leading index that tracks deviations in the Fed's communication towards inflation targeting. We show that deviations in the index shift inflation expectations and impact monetary policy actions, leading to a substantial reduction in the inflation response parameter of an estimated Taylor-Rule.

We view this paper to be just a first step towards answering many exciting questions, such as developing novel measures for sentiment and uncertainty, modeling institutional differences, and improving the accuracy of real-time predictions. We hope that by making our language models publicly available, we will be able to assist in this process.

Declaration of competing interest

The author declares that he has no relevant or material financial interests that relate to the research described in the paper.

Appendix A

See Table A.1.

Table A.1
MP frameworks — Summary statistics.

	N	Mean	St. Dev.	Pctl(25)	Median	Pctl(75)
Main Regression 1: MP Targets						
Cobham's (2021) MP targets						
ITs	957	0.62	0.49	0	1	1
– FIT	957	0.22	0.41	0	0	0
– LIT	957	0.36	0.48	0	0	1
– FCIT	957	0.01	0.09	0	0	0
– LCIT	957	0.03	0.17	0	0	0
LSD	957	0.25	0.43	0	0	0
ERTs	957	0.10	0.30	0	0	0
WSD	957	0.01	0.11	0	0	0
ERfix	957	0.02	0.12	0	0	0
NoNat	957	0.001	0.03	0	0	0
MixedTs	957	0.01	0.08	0	0	0
RBNZ Controls						
Similarity to RBNZ	939	0.39	0.10	0.33	0.39	0.45
Inflation Δ to NZL	939	1.49	4.27	−0.95	0.64	2.79
log(GDP) Δ to NZL	939	−0.91	1.79	−2.02	−0.30	0.42
Unemployment Rate Δ to NZ	939	1.38	5.22	−1.93	0.39	3.67
Fed Controls						
Similarity to Fed	939	0.48	0.09	0.41	0.49	0.55
Inflation Δ to US	939	1.20	4.27	−1.28	0.30	2.75
log(GDP) Δ to US	939	−1.45	1.74	−2.46	−0.89	−0.13
Unemployment Rate Δ to US	939	1.29	5.07	−2.00	0.23	3.31
ECB Controls						
Similarity to ECB	939	0.52	0.11	0.44	0.51	0.58
Inflation Δ to EA	939	1.06	4.60	−1.65	0.27	2.79
log(GDP) Δ to EA	939	7.18	1.43	6.12	7.65	8.35
Unemployment Rate Δ to EA	939	−2.23	5.03	−5.63	−3.21	−0.30
Main Regression 2: Expectations						
IT	241	0.01	1.01	−0.71	0.03	0.70
$E_t[\pi_{1y}] - 2\%$	241	−0.01	0.63	−0.39	−0.05	0.36
$\lvert E_t[\pi_{1y}] - 2\% \rvert$	241	0.49	0.39	0.20	0.39	0.68
$\lvert E_t[\pi_{2y}] - 2\% \rvert$	241	0.44	0.31	0.19	0.39	0.62
$\lvert E_t[\pi_{10y}] - 2\% \rvert$	241	0.35	0.27	0.14	0.31	0.47
$\lvert E_t[\pi_{1y}^{Mich}] - 2\% \rvert$	241	0.97	0.53	0.70	0.90	1.20
$E_t[r_{1m}]$	241	0.11	2.00	−1.28	−0.26	1.25
$E_t[r_{1y}]$	241	0.14	1.53	−1.02	−0.17	1.12
$E_t[r_{10y}]$	241	1.24	0.84	0.55	1.09	1.91
Infl. Risk Premium	241	0.40	0.07	0.38	0.41	0.44
Real Risk Premium	241	1.22	0.06	1.17	1.21	1.25
Inflation Rate - 2%	241	0.17	1.24	−0.46	0.13	1.01
Sentiment (LM, 2016)	241	0.17	0.15	0.08	0.17	0.27
Uncertainty (BBD, 2016)	241	1.61	1.47	0.69	1.23	2.00
Hawk-Dove (BN, 2017)	241	0.44	0.25	0.30	0.47	0.61
FG Shocks (Swanson, 2021)	241	−3.85	80.15	−17.62	0.00	14.78
Main Regression 3: Taylor Rule						
IT	85	−0.01	1.02	−0.80	0.15	0.56
IT (Speaker Fixed Effects)	85	0.04	0.97	−0.63	0.18	0.68
IT (Expanding Window)	85	−0.01	1.02	−0.80	0.15	0.56
Shadow Rate (Wu, Xia, 2016)	85	1.42	2.46	−0.50	1.13	2.83
Shadow Short Rate (Krippner, 2021)	85	1.23	2.67	−0.17	1.01	2.98
Inflation Rate - 2%	85	0.17	1.16	−0.40	0.11	0.92
Unemployment Rate	85	5.78	1.81	4.40	5.30	6.67
Output Gap	85	0.16	1.15	−0.48	0.16	0.95
Uncertainty (BBD, 2016)	85	1.52	0.81	0.95	1.33	1.85
Hawk-Dove (BN, 2017)	85	0.46	0.15	0.38	0.46	0.55
Sentiment (LM, 2016)	85	0.16	0.10	0.10	0.16	0.22
FG Shocks (Swanson, 2021)	82	−2.40	63.46	−40.07	3.01	35.31

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jinteco.2025.104101.

Data availability

Replication Files. (Mendeley Data)

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